National Semiconductor

November 1994

LM4250 Programmable Operational Amplifier

General Description

The LM4250 and LM4250C are extremely versatile programmable monolithic operational amplifiers. A single external master bias current setting resistor programs the input bias current, input offset current, quiescent power consumption, slew rate, input noise, and the gain-bandwidth product. The device is a truly general purpose operational amplifier.

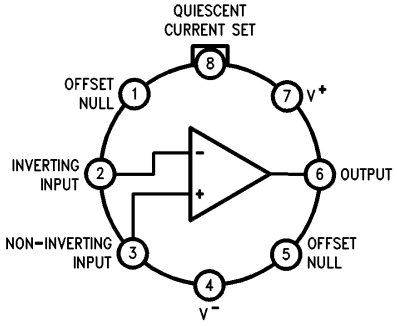
The LM4250C is identical to the LM4250 except that the LM4250C has its performance guaranteed over a 0°C to +70°C temperature range instead of the −55°C to +125°C temperature range of the LM4250.

Features

- ±1V to ±18V power supply operation
- 3 nA input offset current
- Standby power consumption as low as 500 nW
- No frequency compensation required
- Programmable electrical characteristics
- Offset voltage nulling capability
- Can be powered by two flashlight batteries
- Short circuit protection

Connection Diagrams

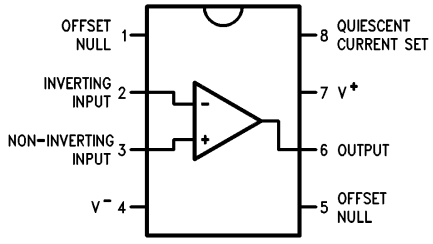
Metal Can Package



TL/H/9300–2

Top View

Dual-In-Line Package



TL/H/9300–5

Top View

Ordering Information

Temperature Range		Package	NSC Package Number
Military $-55°C \leq T_A \leq +125°C$	**Commercial** $0°C \leq T_A \leq +70°C$		
	LM4250CN	8-Pin Molded DIP	N08E
	LM4250CM	8-Pin Surface Mount	M08A
LM4250J LM4250J-MIL		8-Pin Ceramic DIP	J08E
LM4250H LM4250H-MIL	LM4250CH	8-Pin Metal Can	H08C

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Absolute Maximum Ratings

If Military/Aerospace specified devices are required, please contact the National Semiconductor Sales Office/ Distributors for availability and specifications.
(Note 2)

	LM4250	LM4250C
Supply Voltage	±18V	±18V
Operating Temp. Range	$-55°C \leq T_A \leq +125°C$	$0°C \leq T_A \leq +70°C$
Differential Input Voltage	±30V	±30V
Input Voltage (Note 1)	±15V	±15V
I_{SET} Current	150 nA	150 nA
Output Short Circuit Duration	Continuous	Continuous
T_{JMAX}		
H-Package	150°C	100°C
N-Package		100°C
J-Package	150°C	100°C
M-Package		100°C
Power Dissipation at $T_A = 25°C$		
H-Package (Still Air)	500 mW	300 mW
(400 LF/Min Air Flow)	1200 mW	1200 mW
N-Package		500 mW
J-Package	1000 mW	600 mW
M-Package		350 mW
Thermal Resistance (Typical) θ_{JA}		
H-Package (Still Air)	165°C/W	165°C/W
(400 LF/Min Air Flow)	65°C/W	65°C/W
N-Package		130°C/W
J-Package	108°C/W	108°C/W
M-Package		190°C/W
(Typical) θ_{JC}		
H-Package	21°C/W	21°C/W
Storage Temperature Range	−65°C to +150°C	−65°C to +150°C

Soldering Information	
Dual-In-Line Package	
Soldering (10 seconds)	260°C
Small Outline Package	
Vapor Phase (60 seconds)	215°C
Infrared (15 seconds)	220°C

See AN-450 ''Surface Mounting Methods and Their Effect on Product Reliability'' for other methods of soldering surface mount devices.

ESD tolerance (Note 3) 800V

Note 1: For supply voltages less than ±15V, the absolute maximum input voltage is equal to the supply voltage.

Note 2: Refer to RETS4250X for military specifications.

Note 3: Human body model, 1.5 kΩ in series with 100 pF.

Resistor Biasing

Set Current Setting Resistor to V^-

	I_{SET}				
V_S	**0.1 μA**	**0.5 μA**	**1.0 μA**	**5 μA**	**10 μA**
±1.5V	25.6 MΩ	5.04 MΩ	2.5 MΩ	492 kΩ	244 kΩ
±3.0V	55.6 MΩ	11.0 MΩ	5.5 MΩ	1.09 MΩ	544 kΩ
±6.0V	116 MΩ	23.0 MΩ	11.5 MΩ	2.29 MΩ	1.14 MΩ
±9.0V	176 MΩ	35.0 MΩ	17.5 MΩ	3.49 MΩ	1.74 MΩ
±12.0V	236 MΩ	47.0 MΩ	23.5 MΩ	4.69 MΩ	2.34 MΩ
±15.0V	296 MΩ	59.0 MΩ	29.5 MΩ	5.89 MΩ	2.94 MΩ

Electrical Characteristics LM4250 ($-55°C \le T_A \le +125°C$ unless otherwise specified.) $T_A = T_J$

Parameter	Conditions	$V_S = \pm 1.5V$			
		$I_{SET} = 1\ \mu A$		$I_{SET} = 10\ \mu A$	
		Min	Max	Min	Max
V_{OS}	$R_S \le 100\ k\Omega$, $T_A = 25°C$		3 mV		5 mV
I_{OS}	$T_A = 25°C$		3 nA		10 nA
I_{bias}	$T_A = 25°C$		7.5 nA		50 nA
Large Signal Voltage Gain	$R_L = 100\ k\Omega$, $T_A = 25°C$ $V_O = \pm 0.6V$, $R_L = 10\ k\Omega$	40k		50k	
Supply Current	$T_A = 25°C$		7.5 μA		80 μA
Power Consumption	$T_A = 25°C$		23 μW		240 μW
V_{OS}	$R_S \le 100\ k\Omega$		4 mV		6 mV
I_{OS}	$T_A = +125°C$ $T_A = -55°C$		5 nA 3 nA		10 nA 10 nA
I_{bias}			7.5 nA		50 nA
Input Voltage Range		$\pm 0.6V$		$\pm 0.6V$	
Large Signal Voltage Gain	$V_O = \pm 0.5V$, $R_L = 100\ k\Omega$ $R_L = 10\ k\Omega$	30k		30k	
Output Voltage Swing	$R_L = 100\ k\Omega$ $R_L = 10\ k\Omega$	$\pm 0.6V$		$\pm 0.6V$	
Common Mode Rejection Ratio	$R_S \le 10\ k\Omega$	70 dB		70 dB	
Supply Voltage Rejection Ratio	$R_S \le 10\ k\Omega$	76 dB		76 dB	
Supply Current			8 μA		90 μA

Parameter	Conditions	$V_S = \pm 15V$			
		$I_{SET} = 1\ \mu A$		$I_{SET} = 10\ \mu A$	
		Min	Max	Min	Max
V_{OS}	$R_S \le 100\ k\Omega$, $T_A = 25°C$		3 mV		5 mV
I_{OS}	$T_A = 25°C$		3 nA		10 nA
I_{bias}	$T_A = 25°C$		7.5 nA		50 nA
Large Signal Voltage Gain	$R_L = 100\ k\Omega$, $T_A = 25°C$ $V_O = \pm 10V$, $R_L = 10\ k\Omega$	100k		100k	
Supply Current	$T_A = 25°C$		10 μA		90 μA
Power Consumption	$T_A = 25°C$		300 μW		2.7 mW
V_{OS}	$R_S \le 100\ k\Omega$		4 mV		6 mV
I_{OS}	$T_A = +125°C$ $T_A = -55°C$		25 nA 3 nA		25 nA 10 nA
I_{bias}			7.5 nA		50 nA
Input Voltage Range		$\pm 13.5V$		$\pm 13.5V$	
Large Signal Voltage Gain	$V_O = \pm 10V$, $R_L = 100\ k\Omega$ $R_L = 10\ k\Omega$	50k		50k	
Output Voltage Swing	$R_L = 100\ k\Omega$ $R_L = 10\ k\Omega$	$\pm 12V$		$\pm 12V$	
Common Mode Rejection Ratio	$R_S \le 10\ k\Omega$	70 dB		70 dB	
Supply Voltage Rejection Ratio	$R_S \le 10\ k\Omega$	76 dB		76 dB	
Supply Current			11 μA		100 μA
Power Consumption			330 μW		3 mW

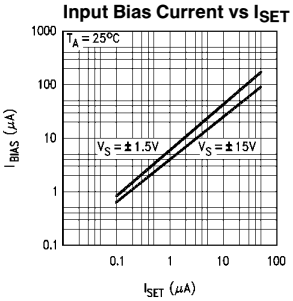
Electrical Characteristics LM4250C ($0°C \le T_A \le +70°C$ unless otherwise specified.) $T_A = T_J$

Parameter	Conditions	$V_S = \pm 1.5V$			
		$I_{SET} = 1\ \mu A$		$I_{SET} = 10\ \mu A$	
		Min	Max	Min	Max
V_{OS}	$R_S \le 100\ k\Omega$, $T_A = 25°C$		5 mV		6 mV
I_{OS}	$T_A = 25°C$		6 nA		20 nA
I_{bias}	$T_A = 25°C$		10 nA		75 nA
Large Signal Voltage Gain	$R_L = 100\ k\Omega$, $T_A = 25°C$ $V_O = \pm 0.6V$, $R_L = 10\ k\Omega$	25k		25k	
Supply Current	$T_A = 25°C$		8 μA		90 μA
Power Consumption	$T_A = 25°C$		24 μW		270 μW
V_{OS}	$R_S \le 10\ k\Omega$		6.5 mV		7.5 mV
I_{OS}			8 nA		25 nA
I_{bias}			10 nA		80 nA
Input Voltage Range		$\pm 0.6V$		$\pm 0.6V$	
Large Signal Voltage Gain	$V_O = \pm 0.5V$, $R_L = 100\ k\Omega$ $R_L = 10\ k\Omega$	25k		25k	
Output Voltage Swing	$R_L = 100\ k\Omega$ $R_L = 10\ k\Omega$	$\pm 0.6V$		$\pm 0.6V$	
Common Mode Rejection Ratio	$R_S \le 10\ k\Omega$	70 dB		70 dB	
Supply Voltage Rejection Ratio	$R_S \le 10\ k\Omega$	74 dB		74 dB	
Supply Current			8 μA		90 μA
Power Consumption			24 μW		270 μW

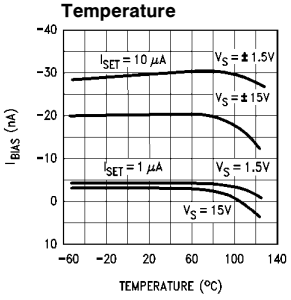
Parameter	Conditions	$V_S = \pm 15V$			
		$I_{SET} = 1\ \mu A$		$I_{SET} = 10\ \mu A$	
		Min	Max	Min	Max
V_{OS}	$R_S \le 100\ k\Omega$, $T_A = 25°C$		5 mV		6 mV
I_{OS}	$T_A = 25°C$		6 nA		20 nA
I_{bias}	$T_A = 25°C$		10 nA		75 nA
Large Signal Voltage Gain	$R_L = 100\ k\Omega$, $T_A = 25°C$ $V_O = \pm 10V$, $R_L = 10\ k\Omega$	60k		60k	
Supply Current	$T_A = 25°C$		11 μA		100 μA
Power Consumption	$T_A = 25°C$		330 μW		3 mW
V_{OS}	$R_S \le 100\ k\Omega$		6.5 mV		7.5 mV
I_{OS}			8 nA		25 nA
I_{bias}			10 nA		80 nA
Input Voltage Range		$\pm 13.5V$		$\pm 13.5V$	
Large Signal Voltage Gain	$V_O = \pm 10V$, $R_L = 100\ k\Omega$ $R_L = 10\ k\Omega$	50k		50k	
Output Voltage Swing	$R_L = 100\ k\Omega$ $R_L = 10\ k\Omega$	$\pm 12V$		$\pm 12V$	
Common Mode Rejection Ratio	$R_S \le 10\ k\Omega$	70 dB		70 dB	
Supply Voltage Rejection Ratio	$R_S \le 10\ k\Omega$	74 dB		74 dB	
Supply Current			11 μA		100 μA
Power Consumption			330 μW		3 mW

Typical Performance Characteristics

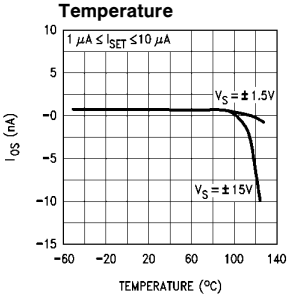
Input Bias Current vs I_{SET}



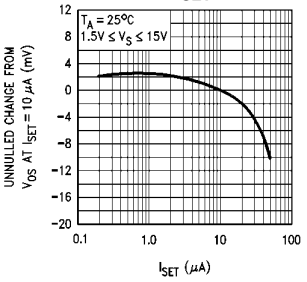
Input Bias Current vs Temperature



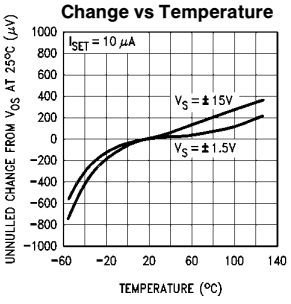
Input Offset Current vs Temperature



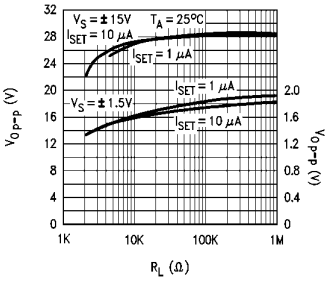
Unnulled Input Offset Voltage Change vs I_{SET}



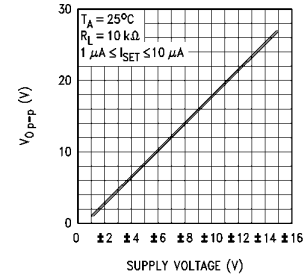
Unnulled Input Offset Voltage Change vs Temperature



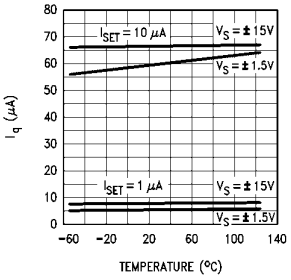
Peak to Peak Output Voltage Swing vs Load Resistance



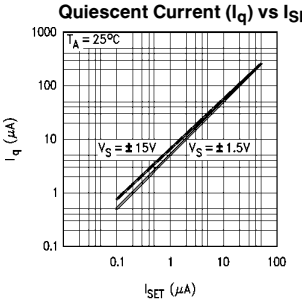
Peak to Peak Output Voltage Swing vs Supply Voltage



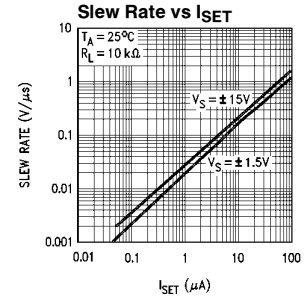
Quiescent Current (I_q) vs Temperature



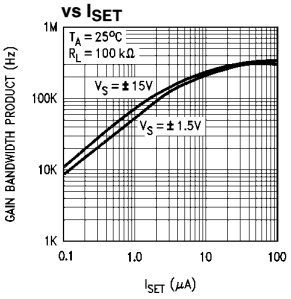
Quiescent Current (I_q) vs I_{SET}



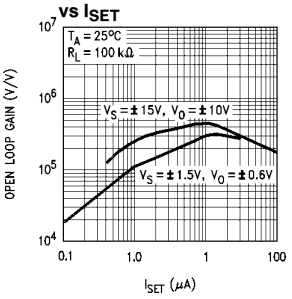
Slew Rate vs I_{SET}



Gain Bandwidth Product vs I_{SET}



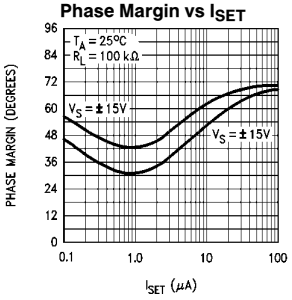
Open Loop Voltage Gain vs I_{SET}



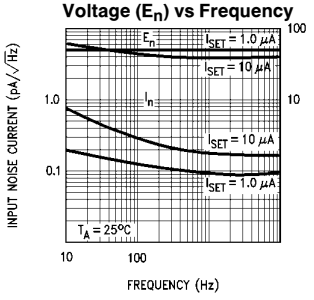
TL/H/9300–6

Typical Performance Characteristics (Continued)

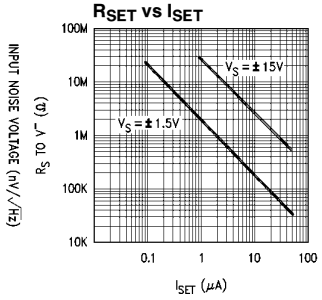
Phase Margin vs I_{SET}



Input Noise Current (I_n) and Voltage (E_n) vs Frequency



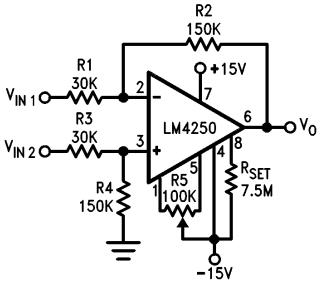
R_{SET} vs I_{SET}



TL/H/9300–7

Typical Applications

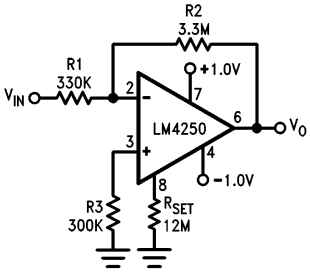
X5 Difference Amplifier



Quiescent P_D = 0.6 mW

TL/H/9300–3

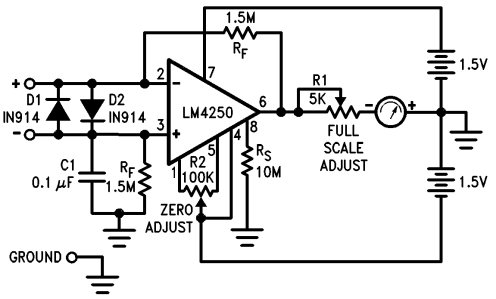
500 Nano-Watt X10 Amplifier



TL/H/9300–4

Quiescent P_D = 500 nW

Floating Input Meter Amplifier
100 nA Full Scale



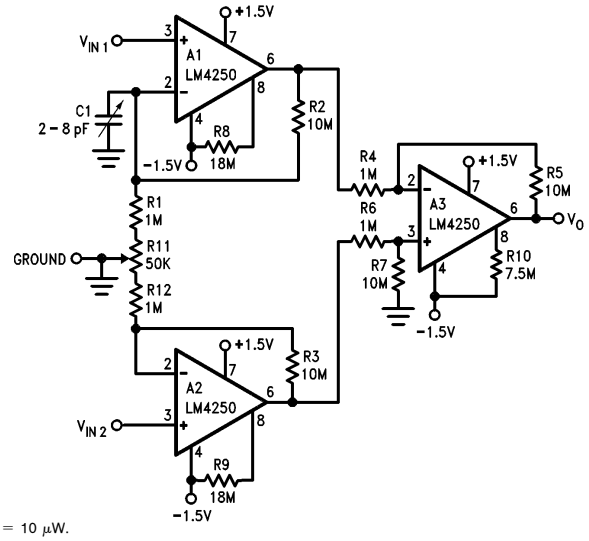
Quiescent P_D = 1.8 μW

*Meter movement (0–100 μA, 2 kΩ) marked for 0–100 nA full scale.

TL/H/9300–8

Typical Applications (Continued)

X100 Instrumentation Amplifier 10 μW



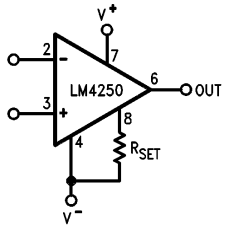
TL/H/9300–9

Note 1: Quiescent $P_D = 10\ \mu W$.

Note 2: R2, R3, R4, R5, R6 and R7 are 1% resistors.

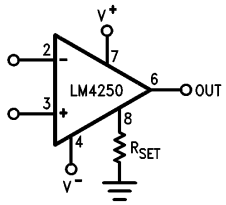
Note 3: R11 and C1 are for DC and AC common mode rejection adjustments.

R_{SET} Connected to V^-



TL/H/9300–10

R_{SET} Connected to Ground



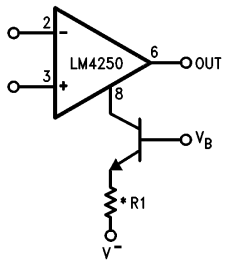
TL/H/9300–11

I_{SET} Equations:

$$I_{SET} \approx \frac{V^+ + |V^-| - 0.5}{R_{SET}}$$ where R_{SET} is connected to V^-.

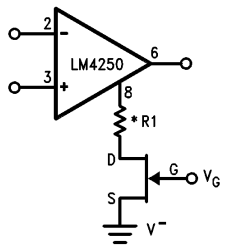
$$I_{SET} \approx \frac{V^+ - 0.5}{R_{SET}}$$ where R_{SET} is connected to ground.

Transistor Current Sourcing Biasing



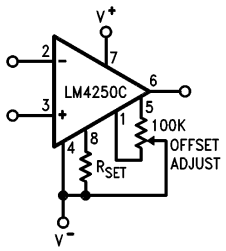
*R1 limits I_{SET} maximum TL/H/9300–12

FET Current Sourcing Biasing



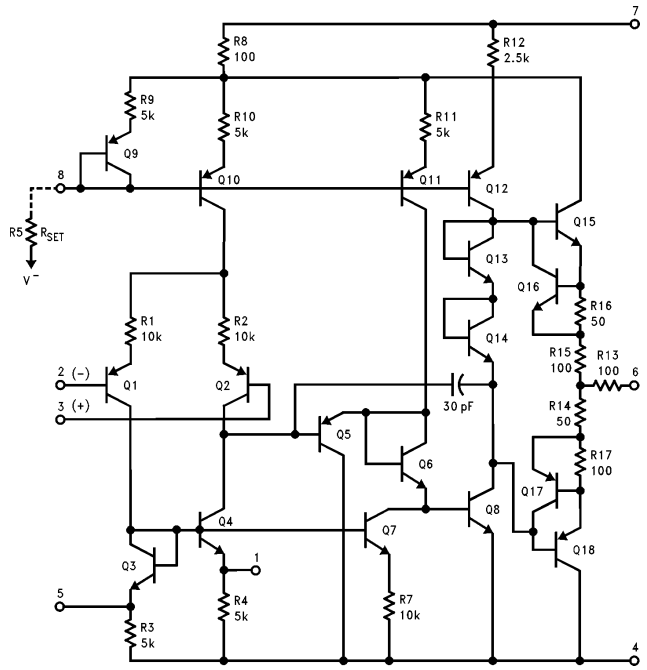
TL/H/9300–13

Offset Null Circuit



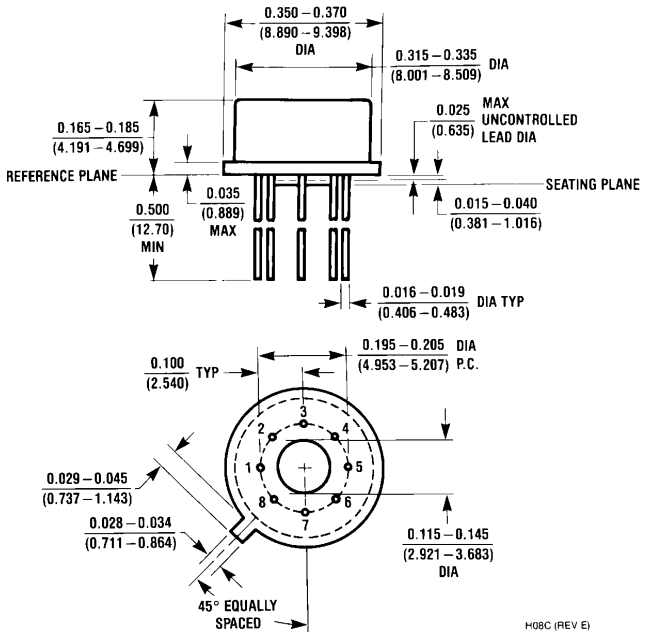
TL/H/9300–14

Schematic Diagram



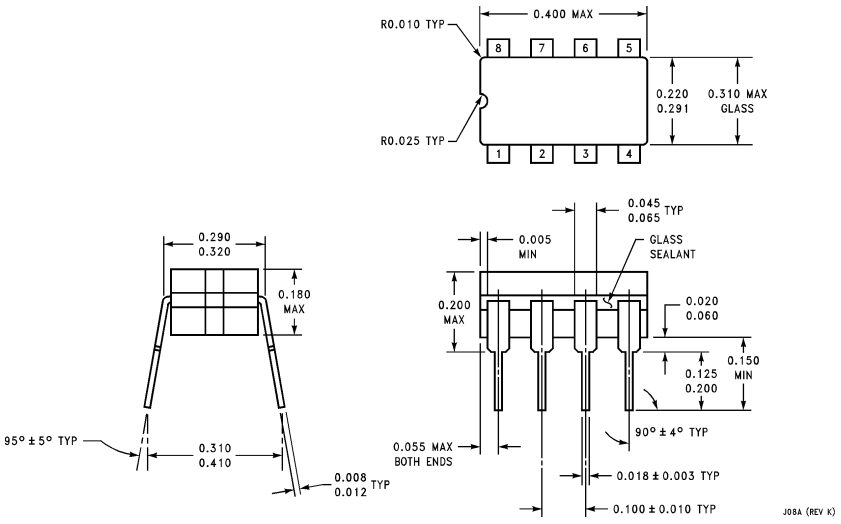
TL/H/9300–1

Physical Dimensions inches (millimeters)

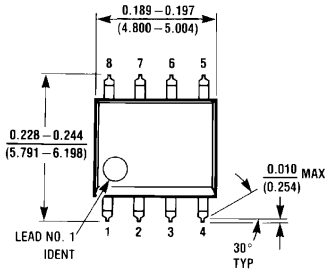


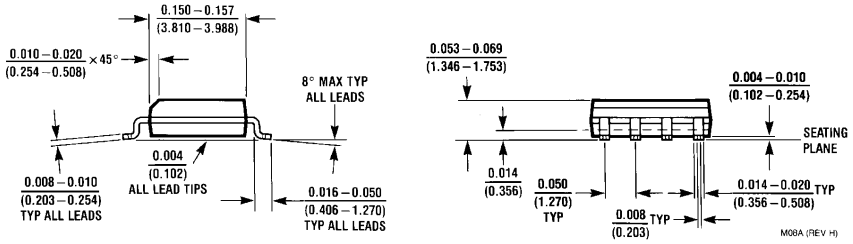
Metal Can Package (H)
Order Number LM4250H, LM4250CH or LM4250H-MIL
NS Package Number H08C

Physical Dimensions inches (millimeters) (Continued)



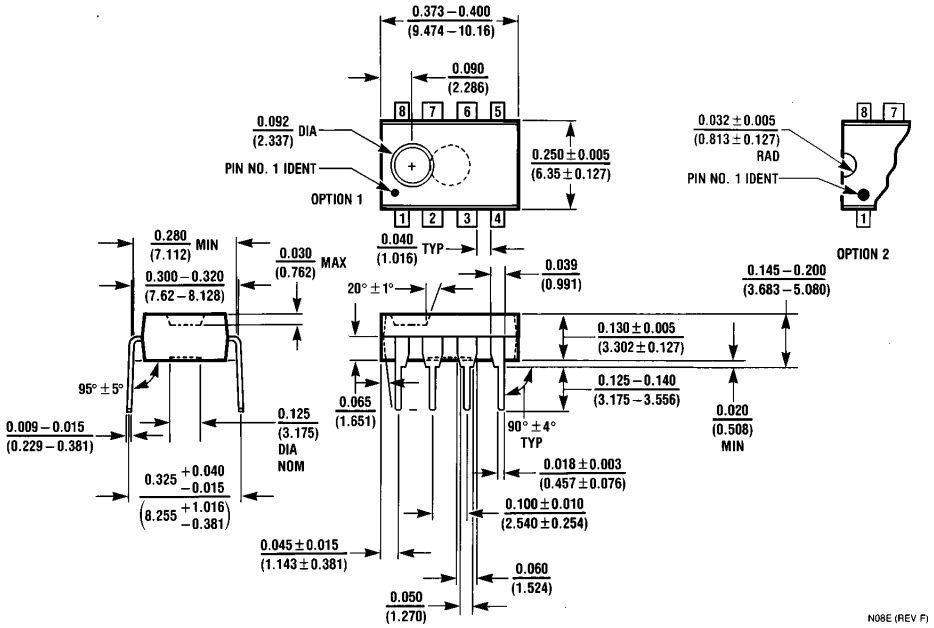
Ceramic Dual-In-Line Package (J)
Order Number LM4250J, or LM4250J-MIL
NS Package Number J08A





Small Outline Package (M)
Order Number LM4250M
NS Package Number M08A

Physical Dimensions inches (millimeters) (Continued)



Molded Dual-In-Line Package (N)
Order Number LM4250CN
NS Package Number N08E

LIFE SUPPORT POLICY

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2. A critical component is any component of a life support device or system whose failure to perform can be reasonably expected to cause the failure of the life support device or system, or to affect its safety or effectiveness.

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